

Amirmahdi Mostofinejad

 amostof |  amirmahdim |  amir.mostofinejad@utoronto.ca |  +1 226 750 4846

SUMMARY

I am a fourth-year Ph.D. candidate actively looking for an internship for summer 2023. My graduate school projects focused on mathematical model development for different biological phenomena, with my master's focusing on tumor-induced angiogenesis (PDE-based) and my Ph.D. on neotissue growth in bioreactors (ODE-based using SciML on Julia). I have also developed a parameter estimator for biology-informed deep learning in SDE-based dynamical systems. My interests are mathematical modeling, optimization, multiphysics, dynamical systems, deep learning, and physics-informed machine learning.

EXPERIENCE

University of Toronto — Research and teaching assistant May 2019 – Dec. 2023

- Modelling the recellularization process in the tracheal scaffolds using optimization and parameter identification supervised by Prof. Cristina Amon
- Developing a methodology for inferring explainable biologically-informed mathematical models and applying it to *in vitro* cell population dynamics using SciML (Julia) and COMSOL Multiphysics

University of Waterloo — Research and teaching assistant Sept. 2016 – May 2019

- Modeled tumor-induced angiogenesis by modeling the migration of the endothelial cells and their interaction with the extracellular matrix, supervised by Prof. Adil Al-Mayah
- Solved a system of partial differential equations describing the biochemical, mechanical, and vasculature using nonlinear finite element analysis in MATLAB

Prestige Land Iran Co. — Intern May 2014 – Aug. 2014

- Sampled and tested the quality of concrete specimens
- Checked the final stage of construction, design calculations, and drawings

SKILLS

Languages	Python, MATLAB, Julia, C/C++
Libraries	SciML, SciPy, scikit-learn, PyTorch, TensorFlow
Numerical solvers	COMSOL Multiphysics, FEniCS, Ansys, Abaqus FEA
Misc.	ParaView, MeshLab, Solidworks, AutoCAD, gmsh, Git, Linux, L ^A T _E X

PROJECTS

Physics-informed deep learning for inference in dynamical systems [Report](#)

Developed a TensorFlow code for parameter estimation of a system of stochastic differential equations using noisy synthetic data.

SIMPLE solver for the lid-driven cavity problem [Report](#)

Developed a Python solver from scratch to solve the Navier-Stokes equations using “SIMPLE” method.

ATP production in cancer cells

Developed a mathematical model for investigating the three different methods for ATP production in cells.

Effects of matrix degradation on pressure-diameter curves in vessels

Modeled multilayer common iliac artery using GOH model. Investigated effects of different matrix densities on the pressure-diameter curve of the common iliac artery.

EDUCATION

Sep 2019 - present Ph.D. (Mechanical Engineering) at **University of Toronto**
Sep 2016 - May 2019 M.A.Sc. (Civil Engineering) at **University of Waterloo**
Sep 2011 - Sep 2015 B.Eng. (Civil Engineering) at **Sharif University of Technology**

HONORS & AWARDS

Barbara and Frank Milligan Graduate Fellowship Oct. 2020
Graduate studies scholarship issued by the MIE department.

William Dunbar Memorial Scholarship Apr. 2020.
Graduate studies scholarship issued by the MIE department.

MIE fellowship May 20219
Entrance fellowship provided by the MIE department.

OGS award July 2018
Graduate studies scholarship issued by the province of Ontario.

PGS award July 2018
President's Graduate Scholarship.

LEADERSHIP

Director of Academic & Professional Development	AMIGAS, UofT
Member of Executive Committee	ISAW, UWaterloo
Vice President of Scientific Activities	Civil Engineering Student Association, Sharif
VP of Entrepreneurship	Green Tomorrow, Sharif
Student-led charity for struggling families	

PUBLICATIONS & PRESENTATIONS

Mostofinejad, Amirmahdi et al. (2021). "In Silico Modeling of Lung Cell Proliferation". In: *Medicine by design Symposium*.

– (2022). "In Silico Modeling Of BEAS-2B Cell Population Dynamics". In: *TERMIS-AM*. Vol. 28, pp. 47–48.

COURSES

Artificial intelligence, Nonlinear modeling of biological systems, Mathematical cell biology, Cancer mechanics, Finite element analysis, Continuum mechanics, Computational fluid mechanics, Probabilistic modeling.

TEACHING

University of Toronto

MIE334, Numerical Methods I (x3),	CSC108, Computer Fundamentals (x4),
MIE1625, Data Science & Quantitative Analysis (x2),	APS105, Intro to Computer Programming (x2),
BME1478, Coding for Biomedical Engineers (x2).	

University of Waterloo

Statics,	Structural Concrete Design I,
Mechanics of Solids II.	

Sharif University of Technology

Concrete Technology and Lab.